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SCHOOL OF AUTONOMOUS SYSTEMS

Self Driving Car Engineer

Syllabus

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Overview

This Nanodegree prepares you to design, build, and optimize self-driving car systems by mastering computer vision, sensor fusion, localization, and control through hands-on projects using C++, Python, and deep learning.

Nanodegree Program | Advanced | 78 hours | 4.2 (45 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

- matplotlib
- Basic calculus
- Intermediate Python
- Finding derivatives
- C++ proficiency
- Object-oriented Python
- Linear algebra
- Python data analysis libraries

You will also need to be able to communicate fluently and professionally **in written and spoken English**.

Skills You'll Learn

Computer Vision

Logistic regression | Model performance metrics | Camera calibration | Image transformations | Gradient descent | Computer vision fluency | Feedforward neural networks | Waymo open dataset | Model evaluation | TensorFlow | Image classification | Backpropagation | Convolutional neural networks | AI business context | Tensorflow object detection API | Linear regression | Tensors | Machine learning pipeline fundamentals | Exploratory data analysis | Pillow | Object detection | Color models | Keras | Yolo algorithm | Deep learning | Overfitting prevention | Color spaces | Data augmentation | Computer vision image enhancement | Computer vision image filtering

Sensor fusion

Cameras | Sensors | Bird's eye view maps | Sae j3016 levels of driving automation standard | Point clouds | Lidar | Object tracking performance metrics | Waymo lidar specifications | Camera measurement models | Extended kalman filters | Object detection | Lidar technical specifications | Object track score | Jacobians | Region-based convolutional neural networks | Computer vision benchmarks | Object tracking | Radar | Object track state | Lidar visualization | Root-mean-square error | Model performance metrics | Multi-target object tracking | Lidar sensors | Sensor visibility reasoning | Coordinate transformation | Convolutional neural networks | Model evaluation | Range imaging | Waymo open dataset | Lidar measurement models | Kalman filters | Feature extraction | Yolo algorithm | Autonomous vehicle sensor selection

Localization

Object motion models | C++ syntax | Bayes' filter for localization | 1d object localization | Bayes' theorem | Point clouds | 2d object localization | Iterative closest point | Law of total probability | Normal distributions transform | Lidar sensors

| Point cloud library | Lidar | C++ control flow | Lidar visualization | Git | C++ classes | Lidar simulator | Cameras | C++ functions | C++ standard library

Planning

Robot behavior planning | Robot motion planning | Eigen library | A* search algorithm | Robot trajectory generation | Automated path planning | Collision detection | Lidar sensors | Velocity profile generation

Control

Control algorithms | Object state estimation | Object perturbations | Mpc controller | Vehicle action | Lidar sensors | Pid controller

Additional Content: Kalman Filters

Radar

Additional Content: Prediction

Waymo open dataset

Additional Content: Control

Cameras

Additional Content: Deep Learning

Lidar

Additional Content: Functional Safety

Lidar

Courses

01. Welcome to Self Driving Car Engineer Nanodegree

1 hour

Welcome to the Self-Driving Car Engineer Nanodegree program! Learn about the Nanodegree experience, as well as hear from Waymo, one of Udacity's partners for the program.

L1	An Introduction to Your Nanodegree Program	
L2	Getting Help	You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content.
L3	Meet Waymo	Hear from Waymo, one of the most cutting-edge autonomous vehicle companies out there! You'll learn about the company as well as about the Waymo Open Dataset, which you'll use in parts of the program.

02. Computer Vision

19 hours

In this course, you will develop critical Machine Learning skills that are commonly leveraged in autonomous vehicle engineering. You will learn about the life cycle of a Machine Learning project, from framing the problem and choosing metrics to training and improving models. This course will focus on the camera sensor and you will learn how to process raw digital images before feeding them into different algorithms, such as neural networks. You will build convolutional neural networks using TensorFlow and learn how to classify and detect objects in images. With this course, you will be exposed to the whole Machine Learning workflow and get a good understanding of the work of a Machine Learning Engineer and how it translates to the autonomous vehicle context.

L1	Introduction to Deep Learning for Computer Vision	Dive into Deep Learning for Computer Vision, learning about its use cases, history, and what you'll build by the end of the course.
L2	The Machine Learning Workflow	Machine learning is more than just building a model - getting each step of the workflow right is crucial.
L3	Sensor and Camera Calibration	Learn how to calibrate your camera to remove distortions for improved perception.
L4	From Linear Regression to Feedforward Neural Networks	Build skills in linear and logistic regression before taking on feedforward neural networks, a type of deep learning.

L5	Image Classification with CNNs	Convolutional networks improve on feedforward networks for areas such as image classification - let's get started building them!
L6	Object Detection in Images	Object detection builds on classification by finding multiple important objects within a single image - find out how!
L7	Project: Object Detection in an Urban Environment	Use the Waymo dataset to detect objects in an urban environment.

03. Sensor fusion

25 hours

Besides cameras, self-driving cars rely on other sensors with complementary measurement principles to improve robustness and reliability, using sensor fusion. You will learn about the lidar sensor, different lidar types, and relevant criteria for sensor selection. Also, you will learn how to detect objects in a 3D lidar point cloud using a deep-learning approach, and then evaluate detection performance using a set of metrics.

In the second half of the course, you will learn how to fuse camera and lidar detections and track objects over time with an Extended Kalman Filter. You will get hands-on experience with multi-target tracking, where you will initialize, update and delete tracks, assign measurements to tracks with data association techniques, and manage several tracks simultaneously.

L1	Introduction to Sensor Fusion and Perception	Get started with sensor fusion and perception, why they are important, and the history of their development in self-driving cars.
L2	The Lidar Sensor	Learn about the lidar sensor, capable of capturing important 3D data in point clouds.
L3	Detecting Objects in Lidar	Detect objects from the 3D data coming in from a lidar sensor.
L4	Project: Mid-Term Project: 3D Object Detection	Use the Waymo dataset to detect 3D objects in the surrounding environment.
L5	Kalman Filters	Learn from the best! Sebastian Thrun will walk you through the usage and concepts of a Kalman Filter using Python.
L6	Extended Kalman Filters	Build an Extended Kalman Filter that's capable of handling data from multiple sources.
L7	Multi-Target Tracking	Get your tracking skills ready for the real world by learning how to track multiple targets simultaneously.
L8	Project: Final Project: Sensor Fusion and Object Tracking	Use the Waymo dataset, along with sensor fusion, to track multiple 3D objects in the surrounding environment.

04. Localization

16 hours

In this course, you will learn all about robotic localization, from one-dimensional motion models up to using three-dimensional point cloud maps obtained from lidar sensors. You'll begin by learning about the bicycle motion model, an approach to use simple motion to estimate location at the next time step, before gathering sensor data. Then, you'll move onto using Markov localization in order to do 1D object tracking. From there, you will learn how to implement two scan matching algorithms, Iterative Closest Point (ICP) and Normal Distributions Transform (NDP), which work with 2D and 3D data. Finally, you will utilize these scan matching algorithms in the Point Cloud Library (PCL) to localize a simulated car with lidar sensing, using a 3D point cloud map obtained from the CARLA simulator.

L1	Introduction to Localization	Meet the team that will guide you through the localization lessons, and learn the intuition behind robotic localization!
L2	C++ Checkpoint	Are you ready to build Kalman Filters with C++? Take these quizzes to find out!
L3	Markov Localization	Learn the math behind localization, as well as how to implement Markov localization in C++.
L4	Creating Scan Matching Algorithms	Learn about and build two scan matching algorithms for localization: Iterative Closest Point (ICP) and Normal Distributions Transforms (NDT).
L5	Utilizing Scan Matching	Learn how to align point clouds with ICP and NDT before leveraging them to localize a self-driving car in a simulated environment!
L6	Project: Scan Matching Localization	Localize a self-driving car within a point cloud from the CARLA simulator with the localization algorithms you learned in previous lessons - how accurate is your algorithm?

05. Planning

11 hours

Path planning routes a vehicle from one point to another, and it handles how to react when emergencies arise. The Mercedes-Benz Vehicle Intelligence team will take you through the three stages of path planning. First, you'll apply model-driven and data-driven approaches to predict how other vehicles on the road will behave. Then you'll construct a finite state machine to decide which of several maneuvers your own vehicle should undertake. Finally, you'll generate a safe and comfortable trajectory to execute that maneuver.

L1	Behavior Planning	Learn how to think about high-level behavior planning in a self-driving car.
L2	Trajectory Generation	Use C++ and the Eigen linear algebra library to build candidate trajectories for the vehicle to follow.

L3 Motion Planning

L4 Project:
Motion Planning and Decision Making for Autonomous Vehicles

06. Control

7 hours

This course will teach you how to control a car once you have a desired trajectory. In other words, how to activate the throttle and the steering wheel of the car to move it following a trajectory described by coordinates. The course will cover the most basic but also the most common controller: the Proportional Integral Derivative or PID controller. You will understand the basic principle of feedback control and how they are used in autonomous driving techniques.

L1 PID Control	Learn about and how to use PID controllers with Sebastian!
L2 Project: Control and Trajectory Tracking for Autonomous Vehicles	

07. Congratulations!

5 minutes

Congratulations on making it through the program!

L1 Congratulations!

08. Additional Content: Kalman Filters

6 hours

Find additional content here on Unscented Kalman Filters, for sensor fusion and tracking.

L1 Unscented Kalman Filters	While Extended Kalman Filters work great for linear motion, real objects rarely move linearly. With Unscented Kalman Filters, you'll be able to accurately track non-linear motion!

09. Additional Content: Prediction

2 hours

Find additional content here on prediction, helping autonomous vehicles to predict how other vehicles and objects might move in the future.

L1	Prediction	Use data from sensor fusion to generate predictions about the likely behavior of moving objects.

10. Additional Content: Control

2 hours

Find additional content here on Vehicles Models and Model Predictive Control, a more advanced form of control.

L1	Vehicle Models	In this lesson, you'll learn about kinematic and dynamic vehicle models. We'll use these later with Model Predictive Control.
L2	Model Predictive Control	In this lesson, you'll learn how to frame the control problem as an optimization problem over time horizons. This is Model Predictive Control!

11. Additional Content: Point Cloud Library

2 hours

L1	[Optional] Intro to PCL	Learn about the Point Cloud Library (PCL). Use a simulation highway environment to explore lidar sensing and generate point clouds.

12. Additional Content: Deep Learning

7 hours

Find additional content on deep learning here, including fully convolutional networks and semantic segmentation for scene understanding, as well as how to improve inference performance from a speed standpoint.

L1	Fully Convolutional Networks	In this lesson you'll learn the motivation for Fully Convolutional Networks and how they are structured.

L2	Scene Understanding	In this lesson you'll be introduced to the problem of Scene Understanding and the role FCNs play.
L3	Inference Performance	In this lesson you'll become familiar with various optimizations in an effort to squeeze every last bit of performance at inference.

13. Additional Content: Functional Safety

6 hours

Find additional content around the field of functional safety for autonomous vehicles and a high level overview of ISO 26262.

L1	Introduction to Functional Safety	You will learn to make safer vehicles using risk evaluation and systems engineering.
L2	Functional Safety: Safety Plan	A functional safety plan is critical to any functional safety project. Here you will learn what goes into a safety plan so that you can document your own.
L3	Functional Safety: Hazard Analysis and Risk Assessment	In a hazard analysis and risk assessment, you will identify vehicular malfunctions and evaluate their risk levels. You can then derive safety goals defining how your vehicle will remain safe.
L4	Functional Safety: Functional Safety Concept	You will derive functional safety requirements from the safety goals and then add extra functionality to the system diagram. Finally you document your work, a part of functional safety.
L5	Functional Safety: Technical Safety Concept	Once you have derived functional safety requirements, you drill down into more detail. In the technical safety concept, you refine your requirements into technical safety requirements.
L6	Functional Safety at the Software and Hardware Levels	The last step in the vehicle safety design phase is to derive hardware and software safety requirements. In this lesson, you will derive these requirements and refine a software system architecture.

14. Autonomous Systems Interview

4 hours

Learn about interviewing for autonomous systems interviews and find plenty of practice questions depending on the role you are looking for!

L1	Project: Autonomous Systems Interview Practice	Start off with some tips on interviewing for an autonomous systems role, then watch how candidates approach their interview questions. Finish off by practicing some questions of your own!

Meet Your Instructors

David Silver

Staff Software Engineer at Kodiak Robotics

David works on Self-driving trucks at Kodiak. As a Staff Software Engineer, David writes planning, control, simulation and mapping software for self-driving trucks.

Thomas Hossler

Sr Deep Learning Engineer

Thomas is originally a geophysicist but his passion for Computer Vision led him to become a Deep Learning engineer at various startups. By creating online courses, he is hoping to make education more accessible. When he is not coding, Thomas can be found in the mountains skiing or climbing.

Antje Muntzinger

Professor of Computer Vision

Antje Muntzinger is a professor of computer vision at Stuttgart University of Applied Sciences, where she teaches AI and applied mathematics. Previously, she was an algorithm developer and tech lead for sensor fusion in the field of automated driving at Mercedes. She holds a PhD in engineering and a diploma in mathematics.

Andreas Haja

Instructor

Andreas Haja is an engineer, educator, and autonomous vehicle enthusiast. Andreas now works as an engineering professor in Germany. Previously, he developed computer vision algorithms and autonomous vehicle prototypes using C++.

Aaron Brown

Senior Software Engineer

Aaron has over 7 years in the autonomous vehicle field, starting as a Udacity content developer then instructor, focusing on Lidar in the Sensor Fusion and Self-Driving Car Nanodegree Program. For the last 5 years, he's been with Mercedes-Benz R&D North America, working in functional testing, Sensor Fusion, and Machine Learning driven by a passion for technological evolution and problem-solving.

Munir Jojo Verge

Lead Autonomous & AI Systems Developer at MITRE

Before MITRE, Munir was a Motion Planning & Decision-Making Manager at Amazon. He also worked for a 2 Self-driving car companies and for WaltDisney Shanghai building TronLightcycle. Munir holds a B.Eng. in Aerospace, a M.S. in Physics, and a M.S. in Space Studies.

Mathilde Badoual

Fifth year PhD student at UC Berkeley

Mathilde has a strong background in optimization and control, including reinforcement learning and has an engineering diploma from the electrical engineering school Supelec, in France. Previously she worked at Tesla in the energy and optimization team.

Why Udacity

Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately

24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices